

# Dr. Julia Hasty

Publications, Patents, and Presentations

julia.hasty@gmail.com — linkedin.com/in/dr-jay

## Patents

- US Patent 8,829,695 — Compact gasifier-genset architecture (2017)
- US Patent 9,476,352 — Compact gasifier-genset architecture (2017)

## Journal Publications

- “Effective landfill gas management strategies for methane control and reuse technology.” *Journal of Renewable and Sustainable Energy*, 7(4), 041511, AIP (2015). ~6 citations.
- “Catalytic Synthesis of Mixed Alcohols Mediated with Nano-Molybdenum Disulfide Microemulsions.” *Fuel*, 164, 339–346 (2016). ~25 citations.
- “Developing Environmentally Benign & Effective Organic Wood Preservatives: Understanding the Biocidal & Non-biocidal Properties of Extractives in Naturally Durable Heartwood.” *Holzforschung* (2008).
- “Determination of the stability constants for the binding of sulfonated morin with Fe<sup>2+</sup>.” *Inorganic Chimica Acta* (2007).

## Selected Presentations

- “A Fixation with Nitrogen — Greenhouse Gas Emissions from Agriculture.” Oral presentation, COP21 Regenerative Agriculture & Land Use Workshop (co-organized by Kiss the Ground, Project Drawdown, and Regeneration International), Paris, France (2015).
- “The Intersection: Energy Meets Biomass, Climate and Agriculture.” Verge/GreenBiz Conference (2016).
- “Conversion of Biomass-derived Synthesis Gas to Mixed Alcohols Catalyzed by Nano-sized Molybdenum Disulfide.” 8th Annual NYS Biotechnology Symposium (2016).
- “Biomass Gasification Technologies.” Oral presentation, Advanced Energy Conference, Albany, NY (2014).
- “Synthesis of Mixed Alcohols from Biomass using Slurried Molybdenum Disulfide Particles.” Poster, Advanced Energy Conference, Albany, NY (2014).
- “Synthesis of Mixed Alcohols From Biomass Using Slurried Nano-Sized Metal Particles.” Poster, AIChE Annual Meeting, San Francisco, CA (2013).
- “The Development of an Economic Modular Process for the Synthesis of Renewable Alcohol Fuels.” Poster, AIChE Annual Meeting, San Francisco, CA (2013).
- “Upgrading of Biomass-derived Bio-oil to Drop-In Replacement Renewable Hydrocarbon Fuels.” NSF-CBERD Annual Meeting, Denver, CO (2011).
- “Biomass to Biofuel: Combined Heat, Power, Fuel & Fertilizer.” West Coast Convergence for Climate Action, Eugene, OR (2008).
- “Synthesis & Characterization of Novel Carbon Based Solid Acid Catalysts for use in Biodiesel Production.” Blue Ridge ACS Regional Meeting, Radford, VA (2006).

## Awards

- Winner, 2018 Water Abundance XPrize (SkySource)

- NSF grant awards (ALL Power Labs, CPWater Tech)
- USAID grant — Liberia electrification, 250 homes during Ebola outbreak
- New York State innovation award (CPWater Tech)
- PowerBridge New York accelerator cohort (NanoSulf)
- Excellence in Teaching — Stony Brook University, Materials Science & Engineering Dept (2013)